

Term 3 Practice Activity: C3 Linear Densities (No integration)

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Evaluated criteria

- C3.

1 Introductory Linear Density

Problem. Let X be a continuous random variable with density

$$f(x) = \frac{x}{18}, \quad 0 \leq x \leq 6,$$

and $f(x) = 0$ otherwise.

Questions/Tasks.

- (a) State the support of X .
- (b) Verify geometrically that f is a valid PDF.
- (c) Compute $P(0 \leq X \leq 3)$.

C3

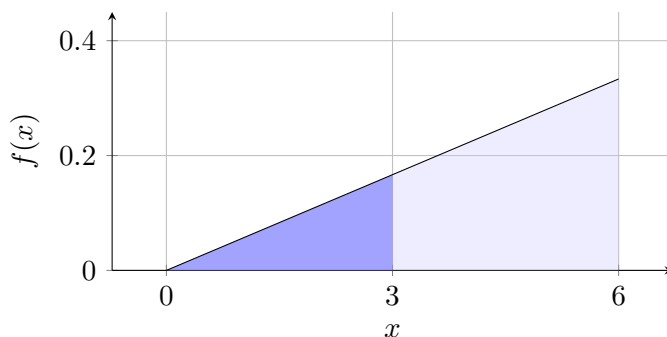
- (a) The support is $[0, 6]$.
- (b) The graph is a single straight line from $(0, 0)$ to $(6, \frac{1}{3})$. So total area is triangle area:

$$\frac{1}{2}(6) \left(\frac{1}{3} \right) = 1.$$

Hence f is valid.

- (c) On $[0, 3]$, the area is a smaller triangle with height $f(3) = \frac{1}{6}$:

$$P(0 \leq X \leq 3) = \frac{1}{2}(3) \left(\frac{1}{6} \right) = \frac{1}{4}.$$



Interpretation. The model increases linearly, so larger values in the support are relatively more likely than very small values.

2 Introductory Linear Density (General Case)

Problem. Let X be a continuous random variable with density

$$f(x) = kx, \quad 0 \leq x \leq b,$$

and $f(x) = 0$ otherwise, where $b > 0$ and $k > 0$.

Questions/Tasks.

- (a) State the support of X .
- (b) Find k geometrically so that f is a valid PDF.
- (c) Compute $P(0 \leq X \leq a)$, where $0 \leq a \leq b$.

C3

- (a) The support is $[0, b]$.
- (b) The graph is a single straight line from $(0, 0)$ to (b, kb) . So total area is triangle area:

$$\frac{1}{2}(b)(kb) = 1.$$

Hence

$$\frac{kb^2}{2} = 1 \quad \Rightarrow \quad k = \frac{2}{b^2}.$$

Therefore,

$$f(x) = \frac{2x}{b^2}, \quad 0 \leq x \leq b.$$

- (c) On $[0, a]$, the area is a smaller triangle with height

$$f(a) = \frac{2a}{b^2}.$$

So

$$P(0 \leq X \leq a) = \frac{1}{2}(a) \left(\frac{2a}{b^2} \right) = \frac{a^2}{b^2}.$$

Interpretation. The model increases linearly, so larger values in the support are relatively more likely than very small values.

3 Basic Symmetric Triangular Density

Problem. A random variable Y has density

$$f(y) = \begin{cases} \frac{y}{16}, & 0 \leq y \leq 4, \\ \frac{8-y}{16}, & 4 \leq y \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Verify this is a valid PDF.
- (b) Compute $P(2 \leq Y \leq 6)$.
- (c) Identify the most likely region and explain why.

C3

- (a) The graph is a triangle with base 8 and peak height $f(4) = \frac{1}{4}$.

$$\text{Area} = \frac{1}{2}(8) \left(\frac{1}{4} \right) = 1.$$

So it is valid.

(b) Use complement:

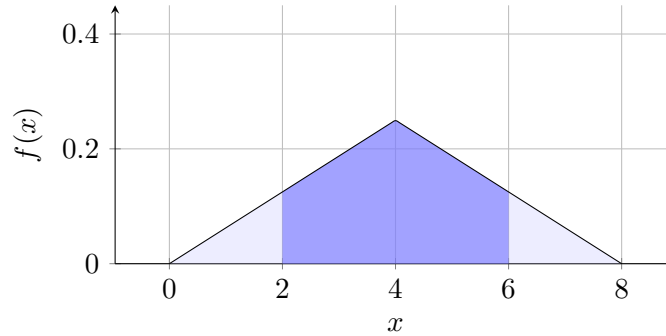
$$P(2 \leq Y \leq 6) = 1 - P(0 \leq Y \leq 2) - P(6 \leq Y \leq 8).$$

Each side area is a triangle with base 2 and height $\frac{1}{8}$:

$$\frac{1}{2}(2) \left(\frac{1}{8} \right) = \frac{1}{8}.$$

Hence

$$P(2 \leq Y \leq 6) = 1 - \frac{1}{8} - \frac{1}{8} = \frac{3}{4}.$$



Interpretation. Values near $y = 4$ are most likely because the density reaches its highest point there.

4 Basic Asymmetric Triangular Density

Problem. A random variable Z has density

$$f(z) = \begin{cases} \frac{z}{20}, & 0 \leq z \leq 4, \\ \frac{10-z}{30}, & 4 \leq z \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Verify the model is a valid PDF.
- (b) Compute $P(1 \leq Z \leq 7)$.
- (c) Where is Z most likely to be found?

C3

(a) Peak at $z = 4$ has height $\frac{1}{5}$. Left triangle area: $\frac{1}{2}(4)(\frac{1}{5}) = \frac{2}{5}$. Right triangle area: $\frac{1}{2}(6)(\frac{1}{5}) = \frac{3}{5}$. Total = 1.

(b) Compute by complement:

$$P(1 \leq Z \leq 7) = 1 - P(0 \leq Z \leq 1) - P(7 \leq Z \leq 10).$$

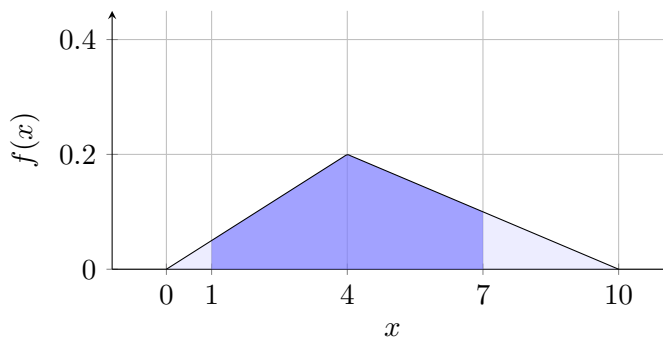
$$P(0 \leq Z \leq 1) = \frac{1}{2}(1) \left(\frac{1}{20} \right) = \frac{1}{40}.$$

At $z = 7$, $f(7) = \frac{1}{10}$, so

$$P(7 \leq Z \leq 10) = \frac{1}{2}(3) \left(\frac{1}{10} \right) = \frac{3}{20}.$$

Therefore

$$P(1 \leq Z \leq 7) = 1 - \frac{1}{40} - \frac{3}{20} = \frac{33}{40} = 0.825.$$



Interpretation. The peak at $z = 4$ marks the most likely region, but the longer right side spreads mass across larger values.

5 Triangular Density with Split-Interval Areas

Problem. Let T have density

$$f(t) = \begin{cases} \frac{t}{9}, & 0 \leq t \leq 3, \\ \frac{6-t}{9}, & 3 \leq t \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Verify validity.
- (b) Compute $P(2 \leq T \leq 5)$ by splitting at the peak.
- (c) Compare $P(1 \leq T \leq 2)$ and $P(4 \leq T \leq 5)$.

C3

- (a) Triangle base 6, height $\frac{1}{3}$ gives area

$$\frac{1}{2}(6) \left(\frac{1}{3} \right) = 1.$$

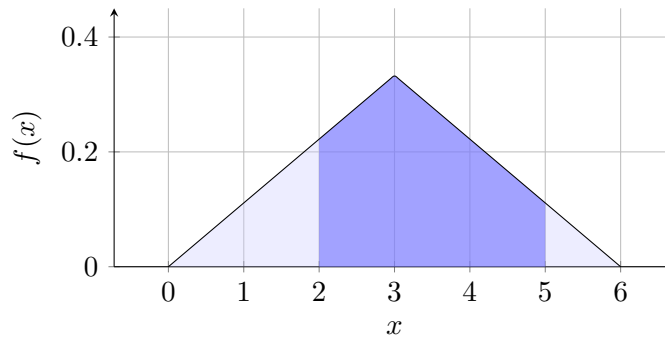
- (b) Split $[2, 5]$ into $[2, 3]$ and $[3, 5]$.

$$A_{[2,3]} = \frac{\left(\frac{2}{9} + \frac{1}{3}\right)}{2} \cdot 1 = \frac{5}{18}, \quad A_{[3,5]} = \frac{\left(\frac{1}{3} + \frac{1}{9}\right)}{2} \cdot 2 = \frac{4}{9}.$$

So

$$P(2 \leq T \leq 5) = \frac{5}{18} + \frac{4}{9} = \frac{13}{18}.$$

- (c) By symmetry around $t = 3$, these two one-unit intervals have equal probability.



Interpretation. The central region around $t = 3$ is most probable because the graph is highest there.

6 Richer Peaked Piecewise Linear Shape

Problem. Let W have density

$$f(w) = \begin{cases} \frac{w}{20}, & 0 \leq w \leq 5, \\ \frac{8-w}{12}, & 5 \leq w \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Verify this is a valid PDF.
- (b) Compute $P(3 \leq W \leq 7)$ by splitting at $w = 5$.
- (c) Which is larger: $P(2 \leq W \leq 4)$ or $P(6 \leq W \leq 8)$? Explain from the graph.

C3

- (a) Peak height is $f(5) = \frac{1}{4}$.

$$\text{Area} = \frac{1}{2}(5) \left(\frac{1}{4} \right) + \frac{1}{2}(3) \left(\frac{1}{4} \right) = \frac{5}{8} + \frac{3}{8} = 1.$$

- (b)

$$A_{[3,5]} = \frac{\left(\frac{3}{20} + \frac{1}{4} \right)}{2} \cdot 2 = \frac{2}{5}, \quad A_{[5,7]} = \frac{\left(\frac{1}{4} + \frac{1}{12} \right)}{2} \cdot 2 = \frac{1}{3}.$$

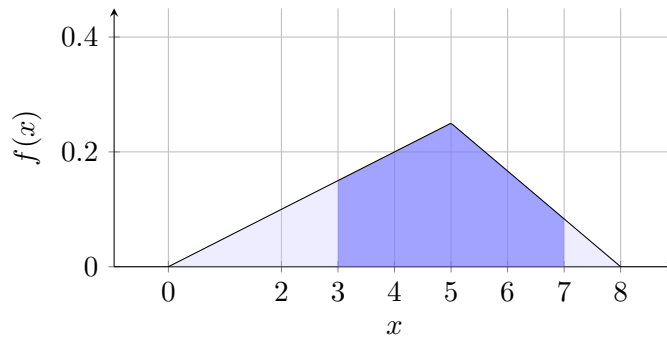
Hence

$$P(3 \leq W \leq 7) = \frac{2}{5} + \frac{1}{3} = \frac{11}{15}.$$

- (c)

$$P(2 \leq W \leq 4) = \frac{\left(\frac{1}{10} + \frac{1}{5} \right)}{2} \cdot 2 = \frac{3}{10}, \quad P(6 \leq W \leq 8) = \frac{1}{2}(2) \left(\frac{1}{6} \right) = \frac{1}{6}.$$

So $P(2 \leq W \leq 4) > P(6 \leq W \leq 8)$.



Interpretation. Because the peak is closer to 5 and the left side is wider, mid-left intervals can hold more area than far-right intervals.

7 Triangular PDF with Unknown Scale k

Problem. A random variable X has density

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 3, \\ k(6 - x), & 3 \leq x \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- Determine k .
- Compute $P(2 \leq X \leq 4)$.
- Explain why the peak corresponds to the most likely region.

C3

- The graph is a symmetric triangle with base 6 and height $3k$:

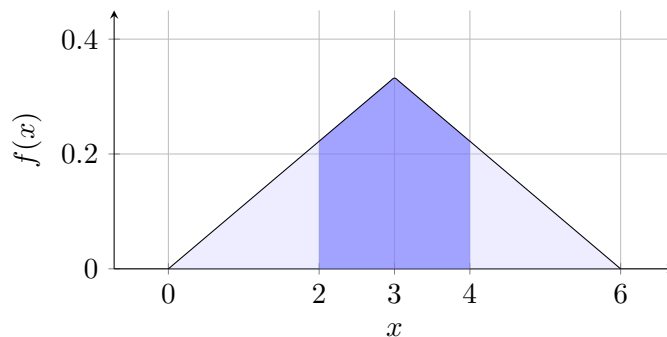
$$\frac{1}{2}(6)(3k) = 9k = 1 \Rightarrow k = \frac{1}{9}.$$

- On $[2, 3]$ and $[3, 4]$:

$$A_{[2,3]} = \frac{\left(\frac{2}{9} + \frac{1}{3}\right)}{2} \cdot 1 = \frac{5}{18}, \quad A_{[3,4]} = \frac{\left(\frac{1}{3} + \frac{2}{9}\right)}{2} \cdot 1 = \frac{5}{18}.$$

So

$$P(2 \leq X \leq 4) = \frac{5}{9}.$$



Interpretation. The density is highest near $x = 3$, so intervals around 3 capture the greatest probability mass.

8 Asymmetric Triangular PDF with Unknown k

Problem. Let Y have density

$$f(y) = \begin{cases} ky, & 0 \leq y \leq 4, \\ \frac{2k}{3}(10 - y), & 4 \leq y \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Find k .
- (b) Compute $P(3 \leq Y \leq 8)$.
- (c) Identify the most likely region from the shape.

C3

(a) Peak is $f(4) = 4k$ (continuous at 4). Total area:

$$\frac{1}{2}(4)(4k) + \frac{1}{2}(6)(4k) = 8k + 12k = 20k = 1 \Rightarrow k = \frac{1}{20}.$$

(b)

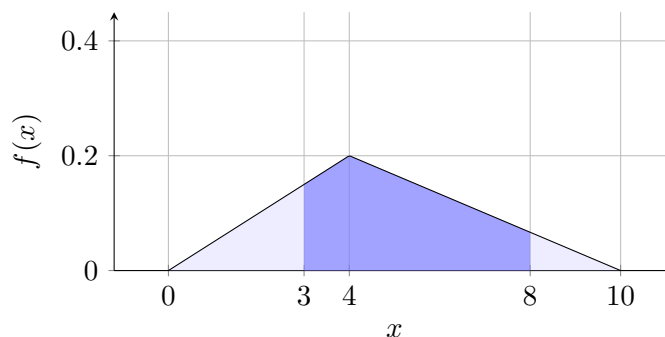
$$A_{[3,4]} = \frac{\left(\frac{3}{20} + \frac{1}{5}\right)}{2} \cdot 1 = \frac{7}{40}.$$

For $[4, 8]$, heights are $\frac{1}{5}$ and $\frac{1}{15}$:

$$A_{[4,8]} = \frac{\left(\frac{1}{5} + \frac{1}{15}\right)}{2} \cdot 4 = \frac{8}{15}.$$

Therefore

$$P(3 \leq Y \leq 8) = \frac{7}{40} + \frac{8}{15} = \frac{17}{24}.$$



Interpretation. The mode is still at $y = 4$, but the longer right tail means probability extends farther to the right.

9 Three-Segment Peaked Density with Parameter

Problem. A variable B on $[0, 12]$ has density

$$f(b) = \begin{cases} \frac{k}{5}b, & 0 \leq b \leq 5, \\ k - \frac{k}{12}(b - 5), & 5 \leq b \leq 9, \\ \frac{2k}{9}(12 - b), & 9 \leq b \leq 12, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Determine k .
- (b) Compute $P(4 \leq B \leq 10)$.
- (c) Explain how the peak location affects where most mass sits.

C3

(a) Areas:

$$A_{[0,5]} = \frac{1}{2}(5)(k) = \frac{5k}{2}, \quad A_{[5,9]} = \frac{(k + \frac{2k}{3})}{2} \cdot 4 = \frac{10k}{3}, \quad A_{[9,12]} = \frac{1}{2}(3) \left(\frac{2k}{3} \right) = k.$$

So

$$\frac{5k}{2} + \frac{10k}{3} + k = \frac{41k}{6} = 1 \Rightarrow k = \frac{6}{41}.$$

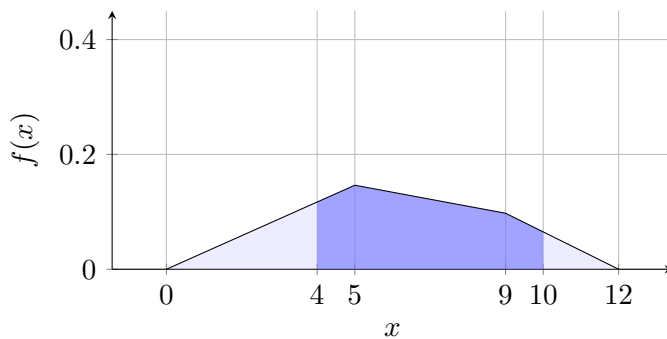
(b) Split into $[4, 5]$, $[5, 9]$, and $[9, 10]$.

$$A_{[4,5]} = \frac{(\frac{4k}{5} + k)}{2} \cdot 1 = \frac{9k}{10}, \quad A_{[5,9]} = \frac{10k}{3},$$

$$A_{[9,10]} = \frac{(\frac{2k}{3} + \frac{4k}{9})}{2} \cdot 1 = \frac{5k}{9}.$$

Hence

$$P(4 \leq B \leq 10) = \frac{9k}{10} + \frac{10k}{3} + \frac{5k}{9} = \frac{431}{615} \approx 0.701.$$



Interpretation. The high central part from about 5 to 9 concentrates much of the probability mass.

10 Comparing Two Triangular-Type Densities

Problem. Two models for waiting time on $[0, 8]$ are:

$$f_A(x) = \begin{cases} \frac{x}{16}, & 0 \leq x \leq 4, \\ \frac{8-x}{16}, & 4 \leq x \leq 8, \end{cases} \quad f_B(x) = \begin{cases} \frac{x}{24}, & 0 \leq x \leq 6, \\ \frac{8-x}{8}, & 6 \leq x \leq 8. \end{cases}$$

Questions/Tasks.

- (a) Show each is a valid PDF.
- (b) Compare $P_A(3 \leq X \leq 5)$ and $P_B(3 \leq X \leq 5)$.
- (c) Explain how moving the peak changes the most likely region.

C3

- (a) Each has base 8 and height $\frac{1}{4}$, so each total area is

$$\frac{1}{2}(8) \left(\frac{1}{4} \right) = 1.$$

- (b) For model A :

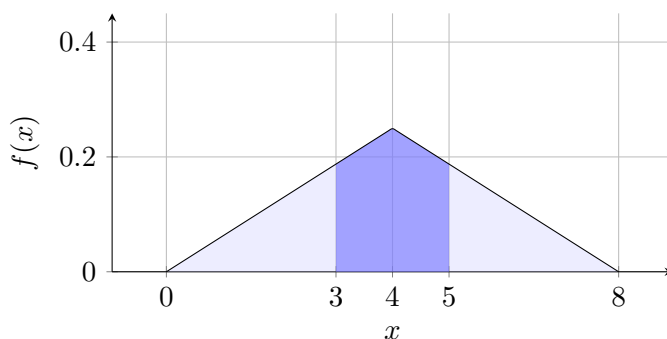
$$P_A(3 \leq X \leq 5) = 2 \cdot \frac{\left(\frac{3}{16} + \frac{1}{4} \right)}{2} \cdot 1 = \frac{7}{16}.$$

For model B , on $[3, 5]$ we are on the rising side only:

$$P_B(3 \leq X \leq 5) = \frac{\left(\frac{1}{8} + \frac{5}{24} \right)}{2} \cdot 2 = \frac{1}{3}.$$

So $P_A(3 \leq X \leq 5) > P_B(3 \leq X \leq 5)$.

- (c) Peak 4 (model A) centers mass near the middle, while peak 6 (model B) shifts more mass to larger values.



Interpretation. Shifting the peak right moves the most likely region right and reduces central-middle probabilities.

11 Variable Peak Location p

Problem. For $0 < p < 10$, define

$$f_p(x) = \begin{cases} \frac{x}{5p}, & 0 \leq x \leq p, \\ \frac{10-x}{5(10-p)}, & p \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Show f_p is a valid PDF for any $0 < p < 10$.
- (b) Express $P(X \leq p)$ in terms of p .
- (c) Explain how changing p affects symmetry and which side carries more probability.

C3

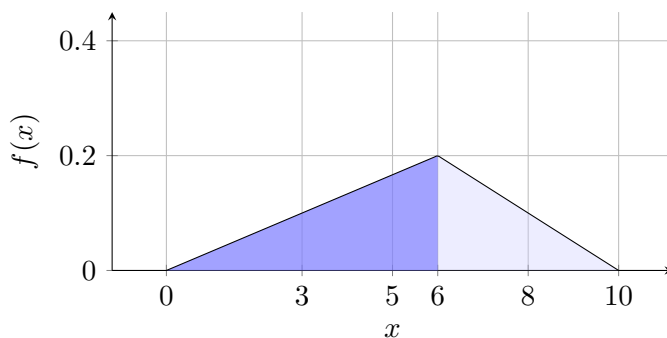
- (a) Peak value is always $f_p(p) = \frac{1}{5}$. Total area is triangle area with base 10 and height $\frac{1}{5}$:

$$\frac{1}{2}(10) \left(\frac{1}{5} \right) = 1.$$

- (b) Left side is a triangle with base p and height $\frac{1}{5}$:

$$P(X \leq p) = \frac{1}{2}p \left(\frac{1}{5} \right) = \frac{p}{10}.$$

- (c) If $p = 5$, the density is symmetric. If $p < 5$, the peak is left-shifted and more mass lies on the right. If $p > 5$, the peak is right-shifted and more mass lies on the left.



Interpretation. The peak location controls which side of the support receives more total area, even though peak height is fixed.

12 Symbolic Height and Asymmetry

Problem. A model on $[0, 8]$ is

$$f(x) = \begin{cases} \frac{h}{3}x, & 0 \leq x \leq 3, \\ h - \frac{h}{5}(x - 3), & 3 \leq x \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

where $h > 0$.

Questions/Tasks.

- (a) Determine h so f is a valid PDF.
- (b) Find $P(2 \leq X \leq 6)$.
- (c) Compare $P(0 \leq X \leq 2)$ with $P(6 \leq X \leq 8)$ and interpret asymmetry.

C3

(a) The graph is triangular with base 8 and height h , so

$$\frac{1}{2}(8)h = 1 \Rightarrow h = \frac{1}{4}.$$

(b) With $h = \frac{1}{4}$:

$$A_{[2,3]} = \frac{(\frac{1}{6} + \frac{1}{4})}{2} \cdot 1 = \frac{5}{24}, \quad A_{[3,6]} = \frac{(\frac{1}{4} + \frac{1}{10})}{2} \cdot 3 = \frac{21}{40}.$$

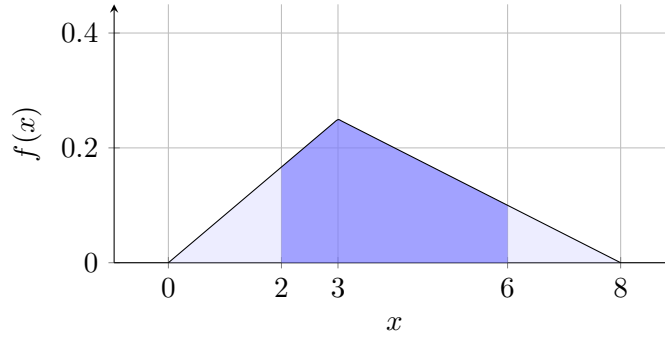
Hence

$$P(2 \leq X \leq 6) = \frac{5}{24} + \frac{21}{40} = \frac{11}{15}.$$

(c)

$$P(0 \leq X \leq 2) = \frac{1}{2}(2) \left(\frac{1}{6} \right) = \frac{1}{6}, \quad P(6 \leq X \leq 8) = \frac{1}{2}(2) \left(\frac{1}{10} \right) = \frac{1}{10}.$$

So $P(0 \leq X \leq 2) > P(6 \leq X \leq 8)$.



Interpretation. Even with one peak, unequal left/right widths allocate probability mass differently.

13 Advanced Parameterized Triangular Model

Problem. For $0 < p < 12$, define

$$f(x) = \begin{cases} kx, & 0 \leq x \leq p, \\ \frac{kp}{12-p}(12-x), & p \leq x \leq 12, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- Determine k in terms of p .
- Show that the peak height is independent of p .
- Find $P(X \leq 6)$ for the case $p = 4$.
- Compare $P(X \leq 6)$ for $p = 4$ and $p = 8$.
- Explain clearly how moving the peak changes the most likely region and concentration of probability mass.

C3

(a) The peak value is $f(p) = kp$. Total area is sum of two triangles:

$$\frac{1}{2}p(kp) + \frac{1}{2}(12-p)(kp) = 6kp = 1 \Rightarrow k = \frac{1}{6p}.$$

(b) Peak height is

$$f(p) = kp = \frac{1}{6},$$

so it does not depend on p .

(c) For $p = 4$, $k = \frac{1}{24}$ and

$$f(x) = \begin{cases} \frac{x}{24}, & 0 \leq x \leq 4, \\ \frac{12-x}{24}, & 4 \leq x \leq 12. \end{cases}$$

Then

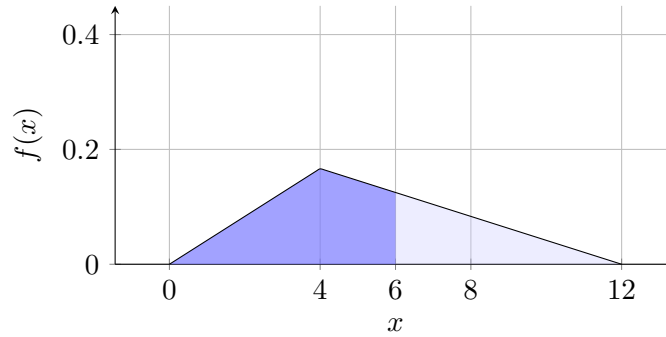
$$P(X \leq 6) = \underbrace{\frac{1}{2}(4) \left(\frac{1}{6}\right)}_{[0,4]} + \underbrace{\frac{(\frac{1}{6} + \frac{1}{6})}{2} \cdot 2}_{[4,6]} = \frac{1}{3} + \frac{7}{24} = \frac{5}{8}.$$

(d) For $p = 8$, by symmetry of the setup around $x = 6$, we get

$$P(X \leq 6) = \frac{3}{8}.$$

So shifting the peak right from 4 to 8 lowers left-half probability.

(e) Moving the peak changes which region is most likely: left-shifted peaks concentrate area earlier, right-shifted peaks concentrate area later.



Interpretation. The highest density marks the most likely region, but interval probability is always determined by area under the piecewise linear graph.

14 First Multi-Parameter Piecewise Linear Density

Problem. Let $0 < q < p$. Define

$$f(x) = \begin{cases} \frac{h}{q}x, & 0 \leq x \leq q, \\ \frac{h}{p-q}(p-x), & q \leq x \leq p, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- (a) Determine h so that f is a valid PDF.
- (b) Find $P(X \leq q)$ in terms of p and q .
- (c) Find $P(\frac{q}{2} \leq X \leq q)$ and $P(q \leq X \leq \frac{p+q}{2})$.
- (d) For fixed p , explain how increasing q changes the shape and concentration of mass.

C3

- (a) The graph is triangular with base p and peak height h , so

$$\frac{1}{2}(p)(h) = 1 \Rightarrow h = \frac{2}{p}.$$

- (b) Left area up to the peak is a triangle:

$$P(X \leq q) = \frac{1}{2}(q)(h) = \frac{1}{2}(q)\left(\frac{2}{p}\right) = \frac{q}{p}.$$

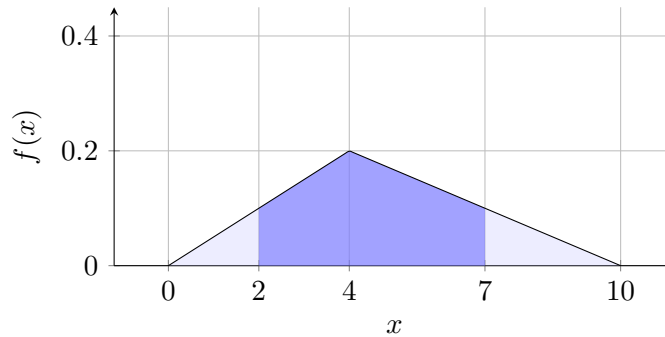
- (c) On $[\frac{q}{2}, q]$, heights are $\frac{h}{2}$ and h :

$$P\left(\frac{q}{2} \leq X \leq q\right) = \frac{(\frac{h}{2} + h)}{2} \cdot \frac{q}{2} = \frac{3hq}{8} = \frac{3q}{4p}.$$

- On $[q, \frac{p+q}{2}]$, heights are h and $\frac{h}{2}$:

$$P\left(q \leq X \leq \frac{p+q}{2}\right) = \frac{(h + \frac{h}{2})}{2} \cdot \frac{p-q}{2} = \frac{3h(p-q)}{8} = \frac{3(p-q)}{4p}.$$

- (d) With p fixed, increasing q moves the peak right. The left side widens and the right side narrows, so a larger fraction of total area lies on the left of the peak ($\frac{q}{p}$ increases).



Interpretation. The peak marks the most likely region, but interval probabilities are still determined by area under the piecewise linear graph.

15 Multi-Parameter Comparison Across Two Settings

Problem. For $0 < q < p$, use the family

$$f_{p,q}(x) = \begin{cases} \frac{2x}{pq}, & 0 \leq x \leq q, \\ \frac{2(p-x)}{p(p-q)}, & q \leq x \leq p, \\ 0, & \text{otherwise.} \end{cases}$$

Consider two parameter settings:

$$\text{Model A: } (p, q) = (10, 4), \quad \text{Model B: } (p, q) = (12, 8).$$

Questions/Tasks.

- Verify that this is a valid PDF for all $0 < q < p$.
- For $0 < r \leq q$, show that $P(X \leq r) = \frac{r^2}{pq}$.
- Compare $P_A(0 \leq X \leq 4)$ and $P_B(0 \leq X \leq 4)$.
- Explain geometrically how changing both p and q changes peak location, asymmetry, spread, and concentration of mass.

C3

- Peak height is

$$f_{p,q}(q) = \frac{2}{p}.$$

Total area is triangular area:

$$\frac{1}{2}(p) \left(\frac{2}{p} \right) = 1,$$

so it is valid.

- If $0 < r \leq q$, the area from 0 to r is a triangle with height $f(r) = \frac{2r}{pq}$:

$$P(X \leq r) = \frac{1}{2}(r) \left(\frac{2r}{pq} \right) = \frac{r^2}{pq}.$$

- Model A: $r = 4 = q$, so

$$P_A(0 \leq X \leq 4) = \frac{4}{10} = \frac{2}{5}.$$

Model B: $r = 4 < q = 8$, so use part (b):

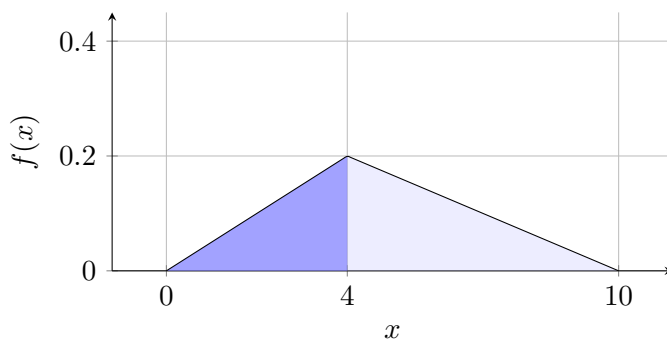
$$P_B(0 \leq X \leq 4) = \frac{4^2}{(12)(8)} = \frac{16}{96} = \frac{1}{6}.$$

Hence

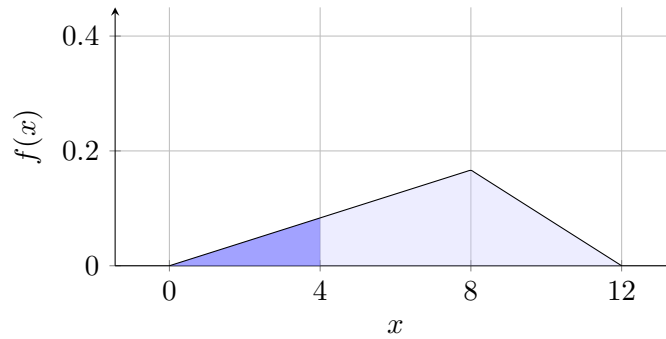
$$P_A(0 \leq X \leq 4) > P_B(0 \leq X \leq 4).$$

- From A to B, support gets wider (p increases) and peak moves farther right (q increases). So the distribution becomes more right-skewed in where mass is concentrated, and early-left intervals (like $[0, 4]$) capture much less area.

Model A graph



Model B graph



Interpretation. Parameter interaction matters: moving the peak and changing support length together can strongly change where probability mass accumulates.

16 Capstone Multi-Parameter Piecewise Linear Model

Problem. Let $a < p < b$ and define

$$f(x) = \begin{cases} C(x-a), & a \leq x \leq p, \\ \frac{C(p-a)}{b-p}(b-x), & p \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Questions/Tasks.

- Determine C in terms of a, p, b .
- Find the peak height $f(p)$ and show how it depends on the support width.
- Find $P(a \leq X \leq p)$.
- Compute $P\left(\frac{a+p}{2} \leq X \leq \frac{p+b}{2}\right)$.
- For $0 < d \leq \min\{p-a, b-p\}$, compare $P(p-d \leq X \leq p)$ and $P(p \leq X \leq p+d)$, and explain how $p-a$ and $b-p$ jointly control which side near the peak carries more mass.

C3

(a) Peak height is

$$h = f(p) = C(p-a).$$

Total area is one triangle with base $b-a$ and height h :

$$\frac{1}{2}(b-a)h = 1 \Rightarrow h = \frac{2}{b-a}.$$

Thus

$$C = \frac{h}{p-a} = \frac{2}{(b-a)(p-a)}.$$

(b) From above,

$$f(p) = h = \frac{2}{b-a},$$

so the peak height depends on total support width $b - a$ (wider support gives lower peak).

(c) Left-of-peak probability is left triangle area:

$$P(a \leq X \leq p) = \frac{1}{2}(p - a)h = \frac{p - a}{b - a}.$$

(d) Split at p . On $[\frac{a+p}{2}, p]$, heights are $\frac{h}{2}$ and h :

$$A_L = \frac{(\frac{h}{2} + h)}{2} \cdot \frac{p - a}{2} = \frac{3h(p - a)}{8}.$$

On $[p, \frac{p+b}{2}]$, heights are h and $\frac{h}{2}$:

$$A_R = \frac{(h + \frac{h}{2})}{2} \cdot \frac{b - p}{2} = \frac{3h(b - p)}{8}.$$

So

$$P\left(\frac{a+p}{2} \leq X \leq \frac{p+b}{2}\right) = A_L + A_R = \frac{3h(b - a)}{8} = \frac{3}{4}.$$

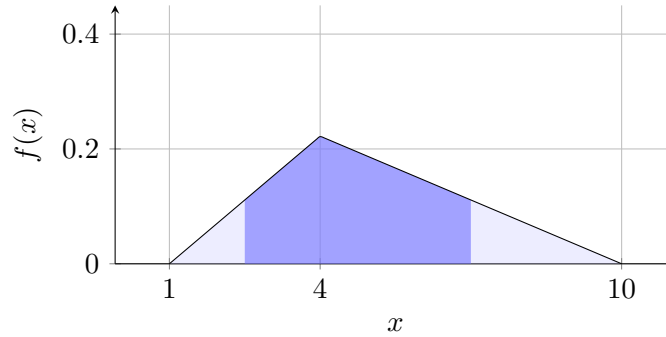
(e) Left near-peak strip:

$$P(p - d \leq X \leq p) = \frac{\left(h\left(1 - \frac{d}{p-a}\right) + h\right)}{2} \cdot d = hd\left(1 - \frac{d}{2(p-a)}\right).$$

Right near-peak strip:

$$P(p \leq X \leq p + d) = \frac{\left(h + h\left(1 - \frac{d}{b-p}\right)\right)}{2} \cdot d = hd\left(1 - \frac{d}{2(b-p)}\right).$$

If $p - a < b - p$, then $\frac{1}{p-a} > \frac{1}{b-p}$ so the left expression is smaller; more near-peak mass is on the right. If $p - a > b - p$, the opposite occurs. If equal, the two probabilities are equal.



Interpretation. This model shows that two shape parameters ($p - a$ and $b - p$) jointly control asymmetry: the most likely region is around p , but nearby left/right interval probabilities differ according to how steeply each side rises or falls.